# 🔹 DSA Week-wise 25% Progress Plan (First 2 Months)

**Goal:** By the end of 8 weeks, **25% of DSA topics** (basics → intermediate) completed, with visible coding and mini-projects.

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## Week 1 – Java Basics & Arrays Introduction ✅ Easy

**Topics:**

- Java syntax review: loops, conditionals, functions
- Arrays: 1D array basics, declaration, initialization, traversal
- Time complexity: O(1), O(n) basics

**Coding / Exercises:**

- Print arrays, sum elements, find max/min
- Reverse an array
- Simple two-pointer problems

**Mini-project:** Array-based calculator (sum, max, min, reverse)

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## Week 2 – Advanced Array Problems ✅ Easy → Moderate

**Topics:**

- 2D arrays
- Insertion & deletion in arrays
- Searching: linear search, basic binary search
- Prefix sum arrays

**Coding / Exercises:**

- Row-wise/column-wise traversal of 2D array
- Find element in 2D array
- Prefix sum computation, subarray sums

**Mini-project:** Matrix addition/subtraction tool

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## Week 3 – Strings & Basic Operations ✅ Moderate

**Topics:**

- String operations: length, substring, concatenation
- Palindrome check
- Character counting, reversing strings

**Coding / Exercises:**

- Reverse a string
- Check palindrome
- Count vowels, consonants, frequency of characters

**Mini-project:** Simple string analyzer (counts, palindrome checker)

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## Week 4 – Recursion Basics ✅ Moderate

**Topics:**

- Recursion: concept & stack
- Factorial, Fibonacci sequence
- Recursive sum of array elements
- Introduction to recursive thinking in problem solving

**Coding / Exercises:**

- Factorial (iterative + recursive)
- Fibonacci sequence
- Sum of array using recursion

**Mini-project:** Recursive calculator for sum/product of array

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## Week 5 – More Recursion & Simple Math Problems ✅ Moderate

**Topics:**

- GCD / LCM using recursion
- Counting digits, sum of digits
- Simple recursive patterns

**Coding / Exercises:**

- Find GCD / LCM
- Sum of digits of a number
- Check if number is prime using recursion

**Mini-project:** Recursive math helper (GCD, LCM, sum of digits)

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## Week 6 – Introduction to Linked Lists ✅ Moderate → Hard

**Topics:**

- Singly linked list basics
- Node creation, insertion, deletion, traversal

**Coding / Exercises:**

- Implement singly linked list from scratch
- Insert node at beginning/end
- Delete node by value

**Mini-project:** Simple contact list using singly linked list

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## Week 7 – Linked List Continued + Stack/Queue Intro ✅ Hard

**Topics:**

- Doubly linked list basics
- Stack & Queue concepts
- Implement stack using array and linked list

**Coding / Exercises:**

- Reverse linked list
- Push, pop operations in stack
- Queue operations (enqueue, dequeue)

**Mini-project:** Stack-based parentheses checker

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## Week 8 – Stack/Queue Applications & Hashing Intro ✅ Hard

**Topics:**

- Queue implementation using array and linked list

- Applications: balanced parentheses, next greater element
- Hashing basics: frequency count, simple hash map logic

**Coding / Exercises:**

- Queue operations
- Frequency of elements in array using hash map
- Simple “first non-repeating character” problem

**Mini-project:** Mini problem-solving toolkit: stack + queue + frequency counter